Vallepu Preethika

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	8.45
SRI CHAITANYA Junior Kalasala	Class XII (TELANGANA BOARD OF INTERMEDIATE EDUCATION)	2024	97.10%
DR KKR'S GOWTHAM CONCEPT HS ECIL X ROAD,MEDCHAL DISTRICT	Class X (BOARD OF SECONDARY EDUCATION,TELANGANA)	2022	9.80%

SKILLS

Programming Languages: C, Python, JavaScript, MySQL, HTML, CSS

Machine Learning: NumPy, Pandas, Scikit-learn, PyTorch, Matplotlib

Developer Tools: Git, Google Colab, MATLAB, LaTeX

Operating Systems: Linux, Windows

PROJECTS

Audio Scene Classification & Multi-Label Sound Detection | *Python* | *PyTorch* | *CNN* | *ResNet18* | *Librosa*

- Built an end-to-end **machine learning and deep learning pipeline** for environmental sound classification using ESC-50, acheiving **83.5% accuracy single-label classification**,**75% F1-score multi-label detection** , and automatic **natural-language scene description generation** from detected sound combinations.
- Developed a **CNN-based environmental sound recognition system** using **Mel spectrograms**, with multi-label detection framework , adaptive per-class thresholding, and structural inference for overlapping sound events.
- Implemented **data preprocessing, feature extraction, model evaluation, and rule-based scene reasoning** to generate human-readable environmental scenarios.

Deep Learning Architectures from Scratch | Numpy, PyTorch

- Implemented **CNN, RNN, LSTM, Transformer Encoder, and State Space Model** forward/backward passes from scratch in **NumPy**, including **custom convolution, multi-head attention, and BPTT**, and validated implementations against **PyTorch**.
- Trained and optimized **PyTorch CNN architectures** on **CIFAR-10** using **SGD, RMSProp, and Adam**, applying **Dropout, Batch Normalization, and hyperparameter tuning** to boost generalization.
- Built an **LSTM-based NLP sentiment classification model** (**~70%** validation accuracy) that outperformed a **vanilla RNN (~50%)**, and analyzed vanishing gradients, attention mechanisms, and computational complexity trade-offs.

Image Classification using Deep Learning | PyTorch (Team Work)

- Designed and trained a **custom Convolutional Neural Network (CNN)** incorporating **Residual Blocks**, **Squeeze-and-Excitation (SE) attention**, and **Batch Normalization** for multi-class image classification.
- Applied **data augmentation**, **AdamW optimization**, hyperparameter tuning, and regularization techniques to improve validation performance and model generalization.
- **Achieved 20th place** in the IIT Hyderabad Deep Learning Image Classification course competition.

Unsupervised Learning Toolkit | Python, Numpy, Matplotlib

- Implemented **Principal Component Analysis (PCA)**, **K-Means**, and **Gaussian Mixture Models (GMM)** from scratch in NumPy, including the **Expectation-Maximization (EM)** algorithm, without using machine learning libraries.

- Developed a **PCA-based face reconstruction pipeline** using Eigenfaces for dimensionality reduction, reconstructing test images across varying numbers of principal components (**k = 10-279**) and analyzing the trade-off between **reconstruction error (MSE)** and compression.
- Compared **hard (K-Means)** and **soft (GMM)** clustering across multiple datasets by evaluating convergence using **Within-Cluster Sum of Squares (WCSS)** and **log-likelihood**, with visualizations of cluster assignments and decision boundaries.

RELEVANT COURSES

Artificial Intelligence : Foundations of Machine Learning (**A**), Deep Learning, AI for Humanity, Programming for AI (**A**), Introduction to Modern AI (**A**), Classical AI

Computer Science: Introduction to Programming, DBMS-1, Data Structures & Applications, Algorithms

Mathematics: Probability & Random Variables, Discrete Mathematics, Vector Calculus, Calculus-I, Calculus-II, Matrix Theory, Differential Equations, Numerical Analysis, Numerical methods for solving ODEs(**A+**)

Other: Linear Systems and Signal Processing, Control Systems

POSITIONS OF RESPONSIBILITY

Teaching Assistant | Probability Course

- Assisted the course instructor in conducting tutorials, clarifying student doubts, and evaluating assignments and quizzes.

EXTRACURRICULAR

- Participated in **National Service Scheme** walkathons and community outreach programs, leading initiatives to promote social awareness and public engagement.